

May 1, 2007

Dear Serre,

Your letter gives me an occasion to try to collect my thoughts. I shall begin by listing some obstacles to grasping the relation between automorphic forms and motives.

A. In order to have a correspondence with reasonable functoriality between representations of $G(F)$ (F local non-archimedean) and representations of the Weil–Deligne group $W'(\overline{F}/F)$, Langlands is led to use the correspondence that I call “unitary” in Antwerp II (forms ..., §3, p. Del42–45). Since functoriality is crucial here, against my inclination, I rally to it. The problem already appears for $\mathrm{GL}(2)$.

Example: if π is an irreducible admissible representation of $\mathrm{GL}(2, F)$, its central character ω_π is a character of F^* , i.e. of $W(\overline{F}/F)$ by class field theory. The dual $G^\vee$ of $\mathrm{GL}(2, F)$ is $\mathrm{GL}(2, \mathbb{C})$ [$\mathbb{C}$, since one is considering complex representations of $\mathrm{GL}(2, F)$], and one wants π to correspond to a conjugacy class of representations

$$\sigma : W'(\overline{F}/F) \longrightarrow \mathrm{GL}(2, \mathbb{C}).$$

One wants a relation between ω_π and $\det(\sigma)$, of the form

$$\omega_\pi = \det(\sigma) \parallel \parallel^k.$$

One also wants to be able to consider a group G isomorphic to $\mathrm{GL}(2, F)$, but for which one has not fixed which was the “obvious” two-dimensional representation: V or $V^\vee$? In other words, one wants compatibility with the outer automorphism τ of $\mathrm{GL}(2, F)$ (which likewise induces one on the dual $G^\vee$): if π corresponds to σ , then $\pi \circ \tau$ (isomorphic to $\pi^\vee$) must correspond to $\tau \circ \sigma$ (isomorphic to $\sigma^\vee$). Since ω_π and $\det(\sigma)$ are thereby replaced by their inverses, this is possible only if $k = 0$.

If χ_1 and χ_2 are characters of F^* , the representation $\chi_1 \oplus \chi_2$ of $W'(\overline{F}/F)$ must, for

$$\chi_1 \chi_2^{-1} \neq \parallel \parallel^{\pm 1},$$

correspond to an induced representation, independent of the order in which χ_1 and χ_2 are taken, and with central character $\chi_1 \chi_2$. These constraints are satisfied by the unitary induction of the character (χ_1, χ_2) of a split torus. Unitary: twist by $\delta^{1/2}$.

The drawback is that, although the notion of irreducible admissible representation π is purely algebraic (one can replace $\mathbb{C}$ by any algebraically closed K of characteristic 0, $G^\vee$ becoming a group over K), the correspondence is not: it uses a square root of q (the number of elements of the residue field). An analyst like Langlands believes that q has one square root better than the other. I do not. This forces me to say that there is not one correspondence between the π 's and the σ 's, but two. In order to name one, one must have chosen a character of F^* which is a square root of $\parallel \parallel$ (with values in K when π is $/K$). I shall suppose the chosen square root is unramified, but this is probably not necessary. In the case of $\mathrm{GL}(n)$, if one replaces one choice of $\parallel \parallel^{1/2}$ by another, the ratio of the two is an ε of order 2, and if π corresponds to σ for one choice, then

$$\pi \cdot \varepsilon(\det)^{(n-1)/2}$$

corresponds to σ for the other. Notice that the central character has not changed.

In the global case, likewise, I would want a correspondence between motives and certain automorphic representations only for F endowed with a character of the idele class group which is a square root of $\parallel \parallel$.

Example: the trivial representation of $\mathrm{GL}(2, F)$ corresponds to the representation

$$\| \|^{1/2} \oplus \| \|^{-1/2}$$

of $W'(\bar{F}/F)$.

Example: a classical automorphic form of weight $2k$ for $\mathrm{GL}(2, \mathbb{Z})$ is a function of a lattice $\Lambda \subset \mathbb{C}$ satisfying, among other things,

$$f(\lambda\Lambda) = \lambda^{-2k} f(\Lambda).$$

The group here is $\mathrm{GL}_{\mathbb{R}}(\mathbb{C})$, and the central character is $\| \|^{2k}$ (or $\| \|^{-2k}$, depending on conventions). The determinant of the corresponding ℓ -adic representation “is” $\| \|^{2k-1}$. This shows that the customary correspondence between modular forms and motives is not compatible with the outer automorphism of $\mathrm{GL}(2)$. If one has endowed $\mathbb{Q}$ with the positive square root of $\| \|$ on $\mathbb{A}^*$, I am forced to say that it is

$$\mathrm{vol}(\mathbb{C}/\Lambda)^{1/2} f$$

that one wants to correspond to a motive. I do not see how one could hope to say anything sensible for a unitary group if the correspondence for $\mathrm{GL}(n)$ were not compatible with outer automorphisms.

Question of sign: the $\| \|^{2k}$ is tied to the following nightmare: if T is a torus, and one lets $T(\mathbb{A})$ act on $T(\mathbb{A})/T(\mathbb{Q})$ by left translations, then on functions on $T(\mathbb{Z})/T(\mathbb{Q})$ by transport of structure, one has

$$[t * f](u) = f(t^{-1}u).$$

A character χ defines the automorphic representation $\mathbb{C}\chi$; the action of $T(\mathbb{A})$ on it is by χ^{-1} .

Complement. For G a reductive group over a field k , there is a canonical morphism

$$G(k) \longrightarrow H^1(\mathrm{Gal}(\bar{k}/k), \mu_2) = k^*/k^{*2},$$

defined as follows. It factors through G^{ad} , is compatible with products, and for $R_{k'/k}(G')$ is deduced, via $N : k'^* \rightarrow k^*$, from its analogue for $G'(k')$. This reduces to the case where G is absolutely simple adjoint. Let G_{sc} be the corresponding simply connected group, and Z its center. The desired morphism is deduced from

$$\gamma : Z \longrightarrow \mu_2,$$

to be defined. The question is geometric, and it is enough to define this γ for G split. Choose a torus $T_{\mathrm{sc}} \subset G_{\mathrm{sc}}$ and a Borel B containing it, and take for γ the restriction of the character

$$\delta_B^{1/2} = \prod \omega_i$$

of T_{sc} to Z . The square of this character factors through the image of T_{sc} in G ; its restriction to Z therefore takes values in μ_2 . Changing B changes $\delta_B^{1/2}$ by a character of the torus of G , and γ is therefore well-defined [another argument: two B 's are conjugate, whereas Z is central].

For arbitrary G , this in fact defines a central extension of G^{ad} by μ_2 , whose pull-back to G one may take:

$$1 \longrightarrow \mu_2 \longrightarrow G_1 \longrightarrow G \longrightarrow 1.$$

For the group $G^{\vee}$, this corresponds to a canonical central element c of order dividing 2. For G adjoint (absolutely simple, to simplify), $G^{\vee}$ is simply connected, and the element in question is

$$\prod_{\alpha > 0} \alpha^{\vee}(-1).$$

If E is an irreducible self-dual representation of $G^\vee$, c acts by ± 1 according as E is orthogonal or symplectic (Bourb. Lie VIII §7 prop.12, p. 152). In fact there is a slightly more precise property than Bourb. loc. cit. which characterizes c . Pin $G^\vee$, and let $P = \mathbb{Z}/2 \ltimes G^\vee$ be the semidirect product for the action of $\mathbb{Z}/2$ which respects the pinning and acts on the Dynkin diagram by the opposition involution. Every irreducible representation E of P is self-dual, c is central in P , and c acts on E by ± 1 according as E is orthogonal or symplectic.

In other words, if one regards the representations of P as super vector spaces, by means of the action of c , and modifies the commutativity of the tensor product accordingly, all representations of P admit an invariant (super)symmetric bilinear form.

This extends to any reductive G and gives c central of order dividing 2 in $G^\vee$. These characters/elements must, as for $\mathrm{GL}(n)$, control the effect of changing $\|\cdot\|^{1/2}$.

B. If f is a classical automorphic form of weight $2k$, for instance on $\mathrm{GL}(2, \mathbb{Z})$, and E is the subfield of $\mathbb{C}$ generated by its coefficients, what corresponds to it is a motive M with coefficients in E , or more concretely a compatible system of E_λ -adic representations of dimension 2. A motive with coefficients in E is a motive M endowed with $E \rightarrow \mathrm{End}(M)$, and there is a coefficient-extension functor $M \mapsto E' \otimes_E M$ for $E \subset E'$. Passing to the limit, one defines the category of motives with coefficients in an algebraic extension $\mathcal{E}$ of $\mathbb{Q}$ by

$$(\mathcal{M} \text{ coeff } \mathcal{E}) = \varinjlim_E \mathrm{ind}(\mathcal{M} \text{ coeff } E \subset \mathcal{E}).$$

I would like to restrict myself, to begin with, to having a correspondence between certain automorphic representations and motivic objects, relative to motives with coefficients in the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ in $\mathbb{C}$. The field $\mathbb{C}$ appears because one considers automorphic representations contained in the space of complex-valued functions on $G(\mathbb{A})/G(F)$.

In the case of function fields, one may instead consider functions with values in an algebraically closed field K of characteristic 0, and one should consider motives with coefficients in the algebraic closure of $\mathbb{Q}$ in K . The story should be invariant under $\mathrm{Aut}(K)$ (which acts on this algebraic closure as well as on $\|\cdot\|^{1/2}$).

In the number-field case, one might hope that the automorphic representations having to do with motives are those to which one can attach Hodge numbers, but this is an inaccessible hope: since one does not understand why there exist Maass forms with a Laplacian eigenvalue other than the good one, λ_0 , corresponding to L -functions of real quadratic fields, how could one show that for $\Delta = \lambda_0$ they are all motivic?

One must not hope that every motivated automorphic representation “with coefficients in E ” corresponds to a motivic object with coefficients in E . If F'/F is Galois with quaternion group, the two-dimensional representation of $\mathrm{Gal}(F'/F)$ gives an automorphic representation “with coefficients in $\mathbb{Q}$ ”, but there is no corresponding motive with coefficients in $\mathbb{Q}$ (only with coefficients in L if L splits the quaternions). Similarly, conjecturally, for abelian surfaces with quaternion algebra of endomorphisms.

C. How is one to understand the semidirect product

$${}^L G = \mathrm{Gal}(F'/F) \ltimes G^\vee$$

introduced by Langlands? In the local non-archimedean case, what plays the role of σ in the preceding explanations is

$$\varphi : W'(\overline{F}/F) \longrightarrow {}^L G$$

compatible with the projections to $\mathrm{Gal}(F'/F)$, and taken up to $G^\vee$ -conjugacy. If I allow myself to replace $G^\vee$, a complex group, by $G^\vee(E_\lambda)$, a group over an extension E_λ of $\mathbb{Q}_\ell$, W' by Gal ,

and φ by a continuous homomorphism (which restores the unipotent part of W'), what one considers is

$$H^1(\mathrm{Gal}(\overline{F}/F), G^\vee(E_\lambda)),$$

in whose definition enters the action of $\mathrm{Gal}(\overline{F}/F)$ on $G^\vee$. What corresponds to π is an isomorphism class of $\mathrm{Gal}(\overline{F}/F)$ -equivariant $G^\vee(E_\lambda)$ -torsors (the equivariance also affecting $G^\vee(E_\lambda)$). When things are said this way, the motivic analogue becomes clear.

To spell it out: $\varphi : \mathrm{Gal}(\overline{F}/F) \rightarrow {}^L G(E_\lambda)$ versus H^1 : let P be the trivial $G^\vee(E_\lambda)$. If Γ is a group of automorphisms of $G^\vee(E_\lambda)$, the group of automorphisms of the system (torsor P under $G^\vee(E_\lambda)$) inducing on $G^\vee(E_\lambda)$ an element of Γ is the semidirect product

$$\Gamma \ltimes G(E_\lambda);$$

γ acts by (γ, γ) , $g \in G(E_\lambda)$ acts by (left translation by g , Id). By this isomorphism, φ gives an equivariant torsor. If one changes the trivialization of the torsor, φ changes by a $G(E_\lambda)$ -inner automorphism.

All this leads me to the following suggestion. Let F be a global field and G reductive over F . Suppose given a square root $\|\cdot\|^{1/2}$ of the character $\|\cdot\|$ of the idele class group. If F is a number field, $\|\cdot\|^{1/2}$ has values in $\mathbb{R}$. If F is a function field and one defines automorphic forms as functions with values in an algebraically closed K of characteristic 0, then $\|\cdot\|^{1/2}$ has values in K .

Regard the dual $G^\vee$ of G as a local system, on $\mathrm{Spec}(F)_{\mathrm{et}}$, of reductive groups over $\mathbb{Q}$. Concretely, such an object is the datum, for each sufficiently large étale extension F' of F , of a reductive group over $\mathbb{Q}$, $G^\vee(F')$, functorial in F' , and in such a way that for $F'' \rightarrow F'$, the morphism $G^\vee(F') \rightarrow G^\vee(F'')$ is an isomorphism; definition of $G^\vee$: say that F'/F is large enough if F' splits G . Starting from $G \otimes_F F'$, one defines its dual $G^\vee$ (split and pinned) as usual. In fact I shall only have to use the corresponding local system $G_{\overline{\mathbb{Q}}}^\vee$ of reductive groups over $\overline{\mathbb{Q}}$, where $\overline{\mathbb{Q}}$ is the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$ (respectively K for a function field). It is not necessary, in order to have $G^\vee(F')$, that F' split F : it suffices that $\mathrm{Gal}(\overline{F}/F')$ act trivially on the Dynkin diagram and that F' split the connected center.

Given this, what should correspond to certain automorphic representations π is an isomorphism class of motivic $G_{\overline{\mathbb{Q}}}^\vee$ -torsors σ , i.e. of objects of the following type: for F'/F large enough that $G^\vee(F')$ be defined, a $\overline{\mathbb{Q}}$ -linear tensor functor

$$\mathrm{Rep}(G_{\overline{\mathbb{Q}}}^\vee(F')) \longrightarrow (\text{motives with coefficients in } \overline{\mathbb{Q}} \text{ over } F'),$$

functorial in F' .

Changing the choice of $\|\cdot\|^{1/2}$. Suppose that $\|\cdot\|^{1/2}$ is replaced by $\varepsilon\|\cdot\|^{1/2}$, where ε is a character of order 2 of the idele class group. We constructed in (A., Complement) an extension

$$1 \longrightarrow \mu_2 \longrightarrow \tilde{G} \longrightarrow G \longrightarrow 1.$$

Push it out by $\mu_2 \hookrightarrow \mathbb{G}_m$:

$$\begin{array}{ccccccc} 1 & \longrightarrow & \mu_2 & \longrightarrow & \tilde{G} & \longrightarrow & G \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \parallel \\ 1 & \longrightarrow & \mathbb{G}_m & \longrightarrow & G_1 & \longrightarrow & G \longrightarrow 1 \\ & & \downarrow z^2 & & \downarrow & & \\ & & \mathbb{G}_m & = & \mathbb{G}_m. & & \end{array}$$

Using this diagram, one defines a character, again denoted ε , of $G(\mathbb{A})$, trivial on $G(F)$: lift g to $g_1 \in G_1(\mathbb{A})$, project to $\mathbb{G}_m(\mathbb{A})$, and apply ε . If, for $\|\cdot\|^{1/2}$, π corresponds to σ , then for $\|\cdot\|^{1/2}\varepsilon$, $\pi\varepsilon$ must correspond to σ .

Changing the choice of $\|\cdot\|^{1/2}$ (bis). We also constructed a canonical element c of $G^\vee$, of order dividing 2. The character ε gives a rank 1 Artin motive L_ε over F , endowed with $L_\varepsilon^{\otimes 2} \simeq 1$. The element c makes it possible to twist σ by L_ε : if σ associates the motive $\sigma(E)$ to the representation E on which c acts by $(-1)^n$, then $\sigma \otimes \varepsilon$ associates to E the motive $\sigma(E) \otimes L_\varepsilon$. If, for $\|\cdot\|^{1/2}$, π corresponds to σ_π , then π must still correspond to a motivic torsor when $\|\cdot\|^{1/2}$ is replaced by $\|\cdot\|^{1/2}\varepsilon$: it must correspond to $\sigma \otimes \varepsilon$.

These two rules should of course be compatible.

Loose ends: (i) For G reductive over F , split over F'/F , it is perhaps more natural to consider

$$[G(\mathbb{A}_{F'})/G(F')]^{\text{Gal}(F'/F)} =: G(\mathbb{A}/F)$$

than $G(\mathbb{A})/G(F)$. In the function-field case, this is required if one wants the points of an algebraic stack. For tori, the character group of $T(\mathbb{A}/F)$ seems more pleasant than that of $T(\mathbb{A})/T(F)$.

(ii) What should one do if π appears in $L(G(\mathbb{A})/G(F))$ with multiplicity > 1 ? Can one hope that the π -isotypic component is canonically a direct sum of representations isomorphic to π (and, in the function-field case, each of these summands containing functions corresponding to ℓ -adic sheaves)? And that each summand gives rise to a σ ?

(iii) How should one characterize σ if its isomorphism class is not uniquely determined by the corresponding local isomorphism classes? If such an indeterminacy is responsible for a multiplicity > 1 as in (ii), how is one to relate π 's in $L(G(\mathbb{A})/G(F))$ to σ 's?

Example: $\text{GL}(2)$. Here, to π and to $\|\cdot\|^{1/2}$, one can associate $\pi|\det|^{1/2}$, and this associated automorphic representation is the same for $(\pi, \|\cdot\|^{1/2})$ and for $(\pi\varepsilon, \|\cdot\|^{1/2}\varepsilon)$ (ε of order 2). This makes it natural to attach a σ to $\pi|\det|^{1/2}$ rather than to $(\pi, \|\cdot\|^{1/2})$. This is the usual practice, although it is not compatible with duality. The same thing can be done for $\text{GL}(2n)$.

I had intended to spell out other examples, but they are delaying this letter too much. They will be for next time.

To be continued . . .

All best,

P. Deligne